Determination of the Composition of an Unknown Binary Liquid Mixture

Aim

To determine the composition of an unknown mixture of two liquids by measuring its refractive index using an Abbe refractometer.

Theory

The refractive index of a liquid is defined as the ratio of the velocity of light in vacuum to its velocity in the liquid:

$$n = \frac{c}{v}$$

where c is the velocity of light in vacuum and v is the velocity of light in the liquid.

The refractive index of a binary liquid mixture depends on its composition, provided that the temperature and wavelength of light remain constant. Therefore, the refractive indices of mixtures having known compositions can be measured and plotted as a calibration graph.

The refractive index of the unknown mixture is measured under the same experimental conditions. Its composition is found from the calibration graph.

Apparatus Required

Abbe refractometer, lens tissue, dropper, standard mixtures of liquids A and B, and unknown liquid mixture.

Procedure

1. Clean the upper and lower prism surfaces of the Abbe refractometer using lens tissue and a suitable solvent.

2. Maintain a constant temperature throughout the experiment and record the temperature.

3. Prepare standard mixtures of liquids A and B with known compositions, such as 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, and 100% of liquid A by volume.

4. Place 2–3 drops of the first standard mixture on the lower prism and close the upper prism carefully. Ensure that a uniform liquid film is formed without air bubbles.

5. Illuminate the prism and observe through the eyepiece.

6. Adjust the compensator until the light-dark boundary becomes sharp and colourless.

7. Adjust the refractometer knob until the light-dark boundary coincides with the crosshair.
8. Record the refractive index. Take three readings and calculate the mean refractive index.
9. Clean the prism surfaces and repeat the procedure for all standard mixtures.
10. Plot a calibration graph with percentage composition of liquid A on the x-axis and mean refractive index on the y-axis.
11. Place the unknown mixture on the prism and measure its refractive index.
12. Locate the refractive index of the unknown on the y-axis of the calibration graph. Draw a horizontal line to meet the calibration curve and then a vertical line to the x-axis.
13. Read the percentage of liquid A from the graph. The percentage of liquid B is calculated using:
$$\%B = 100 - \%A$$

Observation Table

Mixture	Liquid A (% v/v)	Liquid B (% v/v)	Refractive Index, n_D
1	0	100	
2	10	90	
3	20	80	
4	30	70	
5	40	60	
6	50	50	
7	60	40	
8	70	30	
9	80	20	
10	90	10	
11	100	0	
Unknown	–	–	

Result

From the calibration graph:

Composition of liquid A = __________% (v/v)

Composition of liquid B = __________% (v/v)

Precautions

- Keep the temperature constant for all measurements.
- Clean the prism surfaces before measuring each sample.
- Avoid formation of air bubbles between the prism surfaces.
- Ensure that the light-dark boundary is sharp and colourless before taking readings.
- Use identical experimental conditions for standard and unknown mixtures.